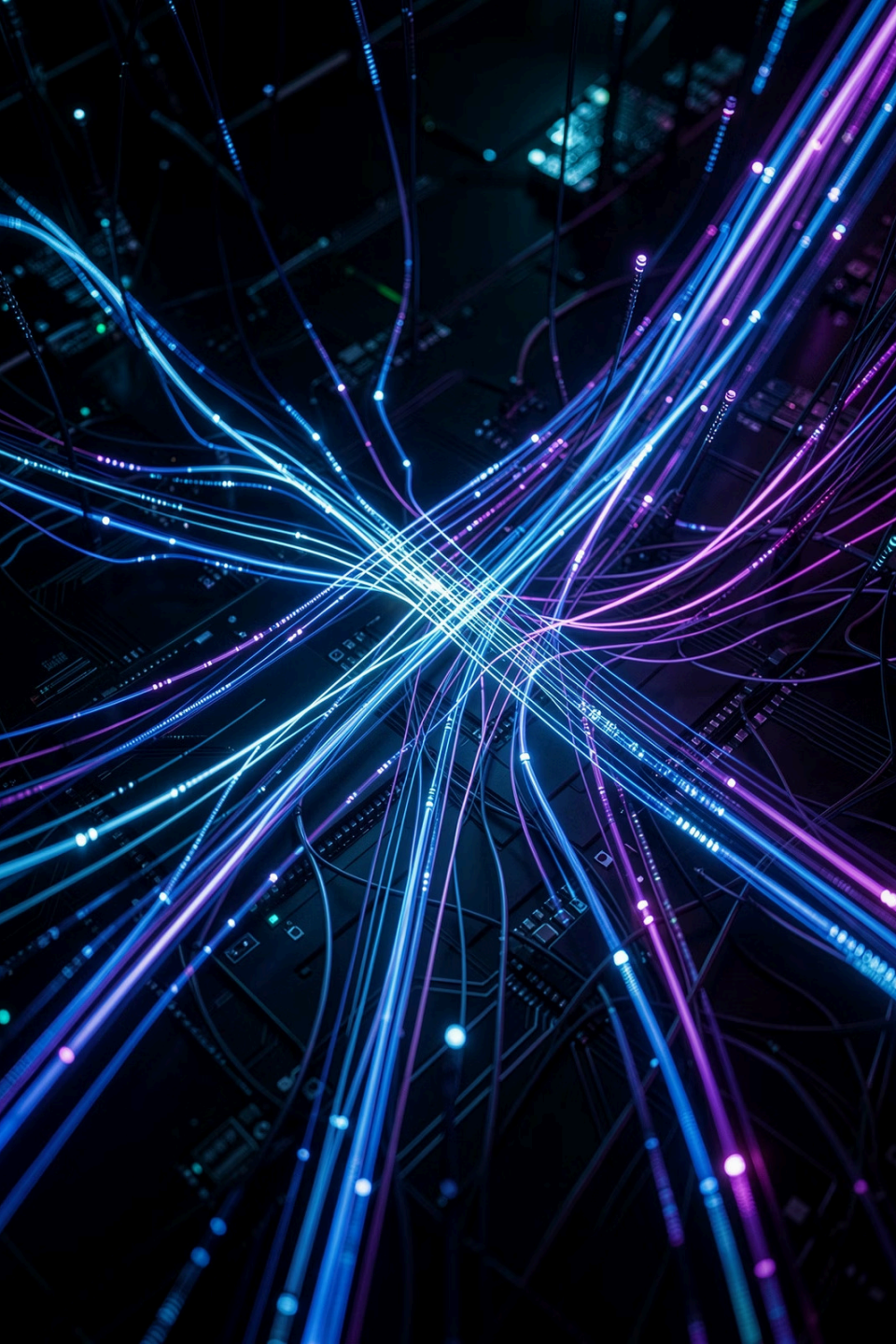




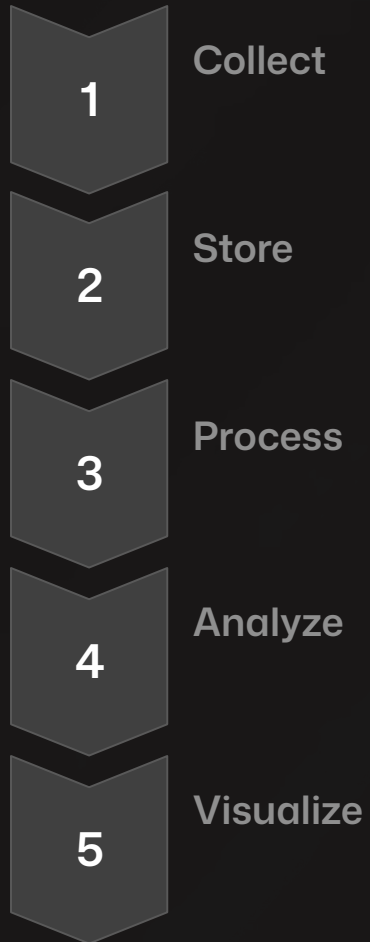
TechPulse

End-to-End Tech News Analytics Platform

A full data analytics project — from scraping tech news to interactive Tableau dashboards. Built to showcase the complete data lifecycle.

Project Objective

TechPulse automatically collects technology news from multiple publishers, stores it in a normalized MySQL database, transforms it into analytical datasets, and presents business insights through SQL, Python, and Tableau.



Technology Stack



Python

Core programming language



MySQL

Relational database



Scraping

Requests, BeautifulSoup, Feedparser



ETL

Python + SQL



Analysis

Pandas, NumPy



Tableau

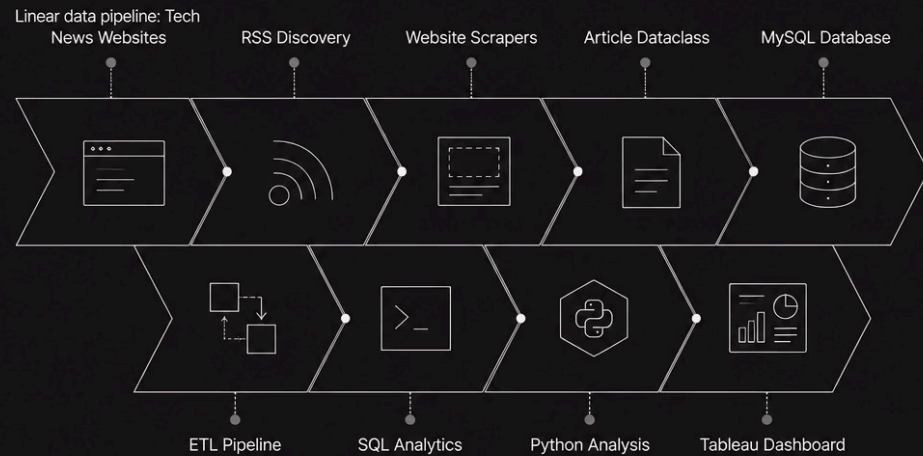
Interactive dashboards



Git & GitHub

Version control

Project Architecture



End-to-End Flow

TechPulse follows a clean, modular architecture. Each stage feeds into the next — from live news sources all the way to polished business intelligence dashboards.

- ❏ Data quality is built incrementally at each stage, not bolted on at the end.

PHASES I - III

Foundation Built

Phase I – Planning

Business requirements, project scope, success metrics, folder structure, and Git repository established.

Phase II – Database Design

Normalized schema with 11 tables including articles, authors, companies, technologies, tags, and bridge tables. ERD and data dictionary completed.

Phase III – Implementation

Tables, constraints, indexes, views, seed data, and validation queries — database is production-ready.



Web Scraper Development



A modular, reusable scraping framework supporting multiple tech news publishers — TechCrunch, The Verge, Ars Technica, and VentureBeat.

01

Article Dataclass

Shared data model

02

BaseScraper

Requests, retries, logging

03

RSS Utility

Discover article URLs

04

Site-Specific Parsers

Per-publisher HTML parsing

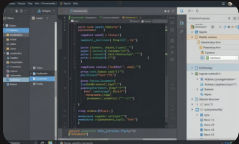
05

Database Loader

Insert articles, authors, tags

Final Deliverables

By the end of TechPulse, you'll have a complete, portfolio-ready data analytics project.



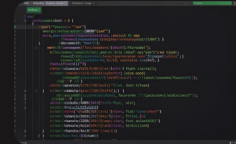
Modular Scraping Framework

Supports multiple tech news websites with reusable components.



Normalized MySQL Database

Proper constraints, indexes, and views.



40-50 SQL Queries

Advanced queries with CTEs, window functions, and ranking.



Tableau Dashboards

Interactive, published to Tableau Public.

Phase IX – X: Documentation & Portfolio



Documentation

README, architecture docs, ERD, data dictionary, methodology, and results.



GitHub Repository

Well-documented with screenshots and GIFs — suitable for data analyst, analytics engineer, or entry-level data engineer portfolios.



Portfolio Presence

Tableau Public, LinkedIn post, resume update, and portfolio website.



TechPulse demonstrates the complete data analytics lifecycle — from raw web data to business intelligence. Follow along as the project unfolds.

